ITSOLERA Pvt. Ltd.
Machine Learning Internship Project Proposal

Project Title: Image Classification Using Traditional Machine Learning Techniques

1. Introduction

Computer vision is a key area of artificial intelligence that enables machines to interpret visual data. While deep learning is widely used for image recognition, understanding traditional machine learning approaches provides a strong foundation.This project focuses on building an **Image Classification System** using classical machine learning techniques. Students will extract features from images and train classifiers to recognize different image categories, gaining practical experience in image preprocessing and feature-based modeling.

2. Objectives

The objectives of this project are:
• To learn the basics of computer vision using traditional ML techniques
• To preprocess and extract features from images
• To train classic machine learning classifiers for image recognition
• To understand the workflow of feature-based image classification

3. Literature Review

Before the deep learning era, classical machine learning approaches such as Support Vector Machines, K-Nearest Neighbors, and Random Forests were widely used for image classification. Feature extraction techniques such as Histogram of Oriented Gradients (HOG), color histograms, and edge detection helped models distinguish between image categories.
Previous studies show that classical methods, while less complex than deep learning, are effective for small datasets and provide insight into fundamental computer vision concepts.

4. Methodology

4.1. Data Collection

We will use a publicly available **image dataset** containing multiple categories. The dataset may include images in formats such as JPEG or PNG and will be used to train and test classifiers.

4.2. Data Preprocessing

• **Resizing:** Standardize image dimensions for consistent feature extraction.
• **Normalization:** Scale pixel values to improve model performance.
• **Grayscale Conversion (optional):** Convert images to grayscale if color is not required.

• **Data Augmentation (optional):** Apply basic transformations such as rotation or flipping to increase dataset size.

4.3. Feature Extraction and Model Training

• **Feature Extraction:** Extract visual features such as edges, shapes, textures, or color histograms.
• **Classifier Selection:** Train traditional machine learning models such as SVM, KNN, or Random Forest.
• **Model Training:** Fit models on the extracted features from the training dataset.

4.4. Model Evaluation

• **Accuracy:** Measure the overall correctness of classification.
• **Precision and Recall:** Evaluate classifier performance for each image category.
• **Confusion Matrix:** Analyze correct and incorrect predictions to identify weaknesses.

5. Expected Outcomes

• An image classification system using traditional machine learning.
• Understanding of image preprocessing and feature extraction techniques.
• Accurate classification of images into categories.
• Strong foundation for transitioning to deep learning-based computer vision projects.

6. Conclusion

This project aims to develop an image classification system using classical machine learning techniques. By preprocessing images, extracting features, and training classifiers, students will gain hands-on experience in computer vision fundamentals. The project provides essential skills for understanding visual data and prepares students for advanced deep learning applications.

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